

Complete measure spaces

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Definition 1. A measure space is complete if any subset of a measure zero set is measurable

- Suppose $(\Omega, \mathcal{F}, \mu)$ is a measure space
- Consider the collection $\mathcal{G}$ of all sets $A \subseteq \Omega$ for which there are $E_1, E_2 \in \mathcal{F}$ satisfying:
 - $\mu(E_2 \setminus E_1) = 0$
 - $E_1 \subseteq A \subseteq E_2$
- Define $\nu : \mathcal{G} \rightarrow [0, \infty]$ by $\nu(A) = \sup\{\mu(E) : E \subseteq A, E \in \mathcal{F}\}$

Claim 1. Suppose $A \in \mathcal{G}$ with E_1, E_2 as before. Then $\nu(A) = \mu(E_1) = \mu(E_2)$

Claim 2. $\mathcal{G}$ is a σ -algebra on Ω

Proof. We show countable unions and complements

- Suppose $A_n \in \mathcal{G}$ with $E_n^1 \subseteq A_n \subseteq E_n^2$ as before. Let $E_1 = \bigcup_{n=1}^{\infty} E_n^1$ and $E_2 = \bigcup_{n=1}^{\infty} E_n^2$. Now

$$E_2 \setminus E_1 = \bigcup_{n=1}^{\infty} (E_n^2 \cap E_1^c) \subseteq \bigcup_{n=1}^{\infty} (E_n^2 \cap (E_n^1)^c)$$

Hence

$$\mu(E_2 \setminus E_1) \leq \sum_{n=1}^{\infty} \mu(E_n^2 \setminus E_n^1) = 0$$

Since $E_1 \subseteq A \subseteq E_2$ where $A = \bigcup_{n=1}^{\infty} A_n$ we get that $A \in \mathcal{G}$

- If $A \in \mathcal{G}$ with E_1, E_2 as before then $E_2^c \subseteq A^c \subseteq E_1^c$. Since $B = E_1^c \setminus E_2^c = E_2 \setminus E_1$ we have $\mu(B) = 0$ which shows closure under complements.

□

Claim 3. ν is a measure on $(\Omega, \mathcal{G})$

Proof. It suffices to prove countable subadditivity and binary additivity:

- Suppose $A_n \in \mathcal{G}$ with E_n^i as before and $B \subseteq A = \bigcup_{n=1}^{\infty} A_n$ with $B \in \mathcal{F}$. We have $B \subseteq \bigcup_{n=1}^{\infty} E_n^2$ so $\mu(B) \leq \sum_{n=1}^{\infty} \mu(E_n^2)$. By the claim we get $\mu(E_n^2) = \nu(A_n)$. Now by the definition of supremum we get countable subadditivity
- Suppose $A_1, A_2 \in \mathcal{G}$ are disjoint with E_1^i, E_2^i as before. Now

$$\nu(A_1 \sqcup A_2) \geq \nu(E_1^1 \sqcup E_2^1) = \mu(E_1^1) + \mu(E_2^1) = \nu(A_1) + \nu(A_2)$$

Finally, use subadditivity as just proved.

□

Claim 4. ν is the unique measure on $(\Omega, \mathcal{G})$ such that $\nu|_{\mathcal{F}} = \mu$

Proof. Assume ν_1 and ν_2 are two such measures. We see that $\nu_i(A) = \nu_i(E_1) + \nu_i(A \setminus E_1)$ by countable additivity. By monotonicity $\nu_i(A \setminus E_1) \leq \nu_i(E_2 \setminus E_1) = 0$. Hence $\nu_i(A) = \mu(E_1)$ □

Claim 5. $(\Omega, \mathcal{G}, \nu)$ is a complete measure space.

Proof. Suppose $\nu(E) = 0$ and $F \subseteq E$. We have E_1, E_2 as before. Since $E_2 \setminus E \subseteq E_2 \setminus E_1$ we have $\nu(E_2 \setminus E) = 0$. By countable additivity $\nu(E_2) = \nu(E_2 \setminus E) + \nu(E) = 0$. Hence $\mu(E_2) = 0$ as well. Since $\emptyset \subseteq F \subseteq E_2$ we get that $F \in \mathcal{G}$ □